## Investigation: MUNDIAL S.A - PRODUTOS DE CONSUMO (CVM code 5312), fiscal year 2014, as it was on 2016-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Têxtil e Vestuário (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2014 DFP, version 1, filed on 2015-04-30; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2015-04-30).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=46282

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −3.041 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 12 (n = 21)
- Percentile in group and year: 20 (n = 370)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.892 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 25 (n = 22)
- Percentile in group and year: 34 (n = 457)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.061 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 66 (n = 22)
- Percentile in group and year: 69 (n = 475)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.945 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 34 (n = 22)
- Percentile in group and year: 27 (n = 465)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 0.928 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 2 (n = 22)

Percentile in group and year: 17 (n = 476)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 0.992 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 69 (n = 21)
Percentile in group and year: 57 (n = 390)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 1.176 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 93 (n = 22)
Percentile in group and year: 80 (n = 469)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.080 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 34 (n = 22)
Percentile in group and year: 33 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 0.997 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 20 (n = 22)
Percentile in group and year: 39 (n = 484)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: −3.4 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 27.9 days)
Percentile in sector and year: 20 (n = 22)
Percentile in group and year: 34 (n = 440)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 30.03 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 79 (n = 21)
Percentile in group and year: 89 (n = 434)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.075 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 34 (n = 22)
Percentile in group and year: 37 (n = 488)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: −1.441 (index)
Components: X1 = −0.185; X2 = −0.094; X3 = 0.010; X4 = 0.011 (indices)
Percentile in sector and year: 31 (n = 24)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 3/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 1
- Change in ROA: 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage: 0
- Change in current ratio: 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin: 0
- Change in asset turnover: 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

## What was known on the date

- DFP 2010, v1: 2011-04-04
- DFP 2010, v2: 2011-04-04
- DFP 2011, v1: 2012-03-31
- DFP 2011, v2: 2012-04-02
- DFP 2011, v3: 2012-04-09
- DFP 2012, v1: 2013-03-29
- DFP 2012, v2: 2013-04-03
- DFP 2012, v3: 2013-04-09

- FCA 2014, v1: 2014-01-14
- DFP 2013, v1: 2014-04-01
- FCA 2014, v2: 2014-05-08
- DFP 2014, v1: 2015-04-30
- DFP 2015, v1: 2016-04-20

DFP and FCA filed with the CVM by the reference date; ITR comes in a later phase.
Material facts and market notices filed by the date (IPE open data, since 2010): 8 material fact documents and 71 market notice documents.
Material facts filed in the 24 months before the date (from 2014-05-01 to 2016-04-30):
- 2016-04-25: Cancelamento de Registro de Companhia Aberta
- 2015-12-21: Cancelamento de Registro de Companhia Aberta
- 2015-11-26: Cancelamento de Registro de Companhia Aberta (version 2)

Older material facts, the market notices and the link to each document are in the JSON and on screen.
Material facts and market notices filed with the CVM by the date, from the IPE open data (since 2010): filing date, category, subject (as filed, in Portuguese) and link. The content of the documents is not downloaded, read or summarized, and the list generates no alert, score or ranking. The counts are of documents: versions, resubmissions and English versions count separately; in the list, versions of the same document and documents with the same date, category and subject appear on a single line, with a link to each one.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Com Ressalva (qualified).
no opinion excerpt was found in the text
Auditor in the 2014 FCA: KPMG Auditores Independentes.
Auditor in the 2015 FCA: KPMG Auditores Independentes.
Auditor in the 2016 FCA: KPMG Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):
- 2012: Directa Auditores -> KPMG Auditores Independentes (transition: two firms in the 2012 FCA). The outgoing firm audited 3 consecutive fiscal years (2009 to 2011), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 22 held-out cases, a single measurement with the frozen reader (dd688da): precision 0.973, coverage 0.915, literal check on 146 of 146 excerpts, no natural-person name (docs/leitor_medida_f1_2.json).

Explanatory notes · Related parties
Note: 11 Partes relacionadas (pages 18 to 22)
- [subsidiaries; other related parties] “como as transações que influenciaram os resultados dos exercícios, relativos a operações com partes relacionadas, decorrem de transações entre entidades controladas e outras partes relacionadas.” (p. 18, original excerpt in Portuguese)
- [parent company; subsidiaries] “Todas as transações entre as empresas controladora e

controladas foram eliminadas nas demonstrações financeiras consolidadas.” (p. 18, original excerpt in Portuguese)
- [parent company; subsidiaries; associates] “Os impactos das transações entre controladora, coligadas e controladas estão demonstrados a seguir:” (p. 18, original excerpt in Portuguese)
- [parent company] “Saldos patrimoniais Debêntures Contas a receber Ativo por mútuo e Passivo por mútuo Controladora por vendas conta corrente e conta corrente Outras” (p. 18, original excerpt in Portuguese)
- Excerpt omitted; see page 18 of the document
- [other related parties] “Zhepar Participações Ltda. - - - - 2.249” (p. 18, original excerpt in Portuguese)
- [other related parties] “Mundial Co - 273 - - -” (p. 18, original excerpt in Portuguese)
- [parent company] “Contas a Ativo por mútuo e Passivo por mútuo Debêntures receber por conta corrente e conta corrente Outras vendas Controladora” (p. 19, original excerpt in Portuguese)
- [parent company] “Resultado Venda de Variação Receitas produtos e cambial financeiras Outras serviços Controladora” (p. 20, original excerpt in Portuguese)
- Excerpt omitted; see page 20 of the document
- [parent company] “Mundial Inc. 4.402 (463) - -” (p. 20, original excerpt in Portuguese)
- [parent company] “Eberle Equipamentos e Processos S.A - - - 3.446” (p. 20, original excerpt in Portuguese)
- [parent company] “Laboratório Avamiller de Cosméticos Ltda. - - - -” (p. 20, original excerpt in Portuguese)
- Excerpt omitted; see page 21 of the document
- [subsidiaries] “As demais transações com as empresas controladas referem-se a transferências de numerários na modalidade de conta corrente entre as empresas.” (p. 21, original excerpt in Portuguese)
- [key management personnel] “Honorários da administração O pessoal-chave da administração inclui os conselheiros fiscais, diretores e membros do conselho de administração da Mundial.” (p. 22, original excerpt in Portuguese)
- [key management personnel] “A remuneração paga ou a pagar ao pessoal-chave da administração, nos exercícios findos em 31 de” (p. 22, original excerpt in Portuguese)
- [key management personnel] “Conselho de administração, fiscal e diretores estatutários 2.746 2.628” (p. 22, original excerpt in Portuguese)
- [key management personnel] “Diretores não estatutários 366 349” (p. 22, original excerpt in Portuguese)

Fields of item 18 of CPC 05 (R1)
- [balances] “Os principais saldos de ativos e passivos em 31 de dezembro de 2014 e 31 de dezembro de 2013, assim” (p. 18, original excerpt in Portuguese)
- [amount of the transactions] “As transações comerciais de compra e venda de produtos, matérias-primas e contratação de serviços são realizadas em condições específicas acordadas entre as empresas da Companhia e levam em consideração os volumes de operações, a periodicidade das transações e a segmentação do processo interno de produção dentro do grupo. Tais transações, dadas as suas características específicas, não são comparáveis às transações realizadas com terceiros não relacionados.” (p. 18, original excerpt in Portuguese)
- [amount of the transactions; balances] “Saldos patrimoniais Debêntures Contas a receber Ativo por mútuo e Passivo por mútuo Controladora por vendas conta corrente e conta corrente Outras” (p. 18, original excerpt in Portuguese)
- [balances] “Saldo em 31/12/2014 304 17.301.638 35.462 62.338 6.924” (p. 18, original excerpt in Portuguese)
- [balances] “Saldos patrimoniais” (p. 19, original excerpt in Portuguese)
- [amount of the transactions; balances] “Contas a Ativo por mútuo e Passivo por mútuo Debêntures receber por conta corrente e conta corrente Outras vendas Controladora” (p. 19, original excerpt in Portuguese)
- [balances] “Saldo em 31/12/2013 389.007 11.374 60.442 42.413 7.104” (p. 19, original excerpt in Portuguese)
- [amount of the transactions] “Resultado Venda de Variação Receitas produtos e cambial financeiras Outras serviços Controladora” (p. 20, original excerpt in Portuguese)

- [balances] “Saldo em 31/12/2014 72.010 1.386 - 2.379” (p. 20, original excerpt in Portuguese)
- [amount of the transactions] “Resultado Venda de Variação Receitas produtos e cambial financeiras Outras serviços” (p. 20, original excerpt in Portuguese)
- [balances] “Saldo em 31/12/2013 71.405 338 41.154 7.104” (p. 20, original excerpt in Portuguese)
- [balances] “Saldos patrimoniais” (p. 21, original excerpt in Portuguese)
- [balances] “Saldo ativo por mútuo e conta Consolidado Debêntures corrente” (p. 21, original excerpt in Portuguese)
- [balances] “Saldo em 31/12/14 304.638 2.966” (p. 21, original excerpt in Portuguese)
- [balances] “Saldo em 31/12/13 389.007 21.799” (p. 21, original excerpt in Portuguese)
- [amount of the transactions] “Consolidado Resultado Receitas financeiras” (p. 21, original excerpt in Portuguese)
- Excerpt omitted; see page 21 of the document
- [balances] “Saldo em 31/12/13 41.154” (p. 21, original excerpt in Portuguese)
- [amount of the transactions] “Em 13 de dezembro de 2013 a Companhia adquiriu debêntures emitidas pela Hercules S.A no montante” (p. 21, original excerpt in Portuguese)
- [balances] “de R$ 389.007, que foram subscritas no seu total à vista pelo saldo decorrente conta de mútuo por conta corrente, conforme descrito na nota explicativa 13.” (p. 21, original excerpt in Portuguese)
- [amount of the transactions] “Em novembro de 2014 a Zhepar Participações rescindiu o contrato assinado anteriormente com a Companhia, nos termos clausula 10 do instrumento, ocasião em que procedeu com a baixa para perda do montante.” (p. 21, original excerpt in Portuguese)
- [balances] “A remuneração paga ou a pagar ao pessoal-chave da administração, nos exercícios findos em 31 de” (p. 22, original excerpt in Portuguese)

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: qualified
- Excerpt omitted; see the auditor's report in the document
- [other matters] “Outros assuntos Demonstrações do valor adicionado Examinamos, também, as demonstrações, individuais e consolidadas, do valor adicionado (DVA), referentes ao exercício findo em 31 de dezembro de 2014, elaboradas sob a responsabilidade da administração da Companhia, cuja apresentação é requerida pela legislação societária brasileira para companhias abertas, e como informação suplementar pelas IFRS que não requerem a apresentação da DVA. Essas demonstrações foram submetidas aos mesmos procedimentos de auditoria descritos anteriormente e, em nossa opinião, exceto pelos possíveis efeitos, se houver, no resultado do exercício de 2014 decorrentes de ajustes de exercícios anteriores do assunto mencionado no parágrafo Base para opinião com ressalva, estão adequadamente apresentadas, em todos os seus aspectos relevantes, em relação às demonstrações financeiras tomadas em conjunto. Demonstrações financeiras em 31 de dezembro de 2013 As demonstrações financeiras de 2013, cujos valores são apresentados para fins de comparabilidade, foram por nós auditados e continham ressalva quanto aos saldos contábeis passivos e de resultado relativos às suas obrigações fiscais, incluindo obrigações sociais.” (cut at the end of a sentence, original excerpt in Portuguese)

Reference Form
Form of 2015, version 6, filed on 2016-04-18.
It is the only version available in the open data; earlier versions are not there.
- Related party (name omitted) · Controlada · Contrato de mútuo com prazo indeterminado. · 39636.16 · Saldo em 31/12/2014 - R$ 39.636,16
- Related party (name omitted) · Controlada · contrato de Mútuo com prazo indeterminado. · 17655978.47 · Em 31/12/2014 - R$17.655.978,47
- Related party (name omitted) · Controlada · Contrato de Mútuo com prazo indeterminado. ·

2687312.83 · Em 31/12/2014 - R$ 2.687.312,83
- Related party (name omitted) · Acionista · Contrato de mútuo com prazo indeterminado · 2248887.07 · 31/12/2014 - R$ 2.248.887,07
- Related party (name omitted) · Coligada · Debêntures a receber · 304637575.93 · em 31/12/2014 - R$ 304.637.575,93
- Related party (name omitted) · Controlada · Contrato de mútuo com prazo indeterminado · 2036369.47 · Em 31/12/2014 - R$ 2.036.369,47
- Related party (name omitted) · Controlada · Contrato de mútuo com prazo indeterminado. · 629380.81 · Em 31/12/2014 - R$ 629.380,81
- Related party (name omitted) · Controlada · Contrato de mútuo com prazo indeterminado. · 9245302.95 · Em 31/12/2014 - R$ 9.245.302,95
- Related party (name omitted) · Controlada · Contrato de Mútuo com prazo indeterminado. · −21159543.54 · Em 31/12/2014 R$ (21.159.543,54)
- Related party (name omitted) · Controlada · Contrato de mútuo com prazo indeterminado. · 613136.63 · em 31/12/2014 - R$ 613.136,63
- Related party (name omitted) · Controlada · Contrato de Mútuo com prazo indeterminado. · −5526942.16 · em 31/12/2014 - R$ (5.526,942,16)
- Related party (name omitted) · Controlada · Contrato de mútuo com prazo indeterminado. · −2269947.98 · Em 31/12/2014 - R$ (2.269.947,98)
- Related party (name omitted) · Controlada · Contrato de Mútuo com prazo indeterminado. · −33342893.15 · Em 31/12/2014 - R$ (33.342.893,15)
- Related party (name omitted) · Controlada · Contrato de Mútuo com prazo indetermiado. · 2594816.38 · Em 31/12/2014 - R$ 2.594.816,38

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, a header line (years, column titles) may still come out as an excerpt: in the reader's measurement, 1 of 19 header texts of the held-out cases came out as excerpts.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-2 (commit dd688da). The generation date is in the demo manifest (demo/manifesto.json).

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: MUNDIAL S.A - PRODUTOS DE CONSUMO.